

Project 5 Scope of Work

Team Members: Anthony Perrotto, Nolan Watkins, Carson Eberhardt, Zane Smith

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Submitted to: Professor Kontar

Project Summary:

Our group has previously completed a risk analysis for Risk Averse, LLC, and they liked our work, so they hired us again to examine a specific natural hazard across 6 different states in one region this time. Natural disasters pose significant risks to communities across the United States, impacting infrastructure, economies, and vulnerable populations. To assess these risks, the Federal Emergency Management Agency (FEMA) developed the National Risk Index (NRI), which evaluates risk using Expected Annual Loss, Social Vulnerability, and Community Resilience. We will use the NRI risk definitions and the NRI census tract data for analysis. Risk Averse, LLC also wants us to tract data and information about impacted populations by natural hazards. The project also looked at how population and community vulnerability may affect the impact of natural disasters. Maps and graphs were used to clearly show patterns and make the information easier to understand.

Project Goals:

1. The goals for this project are to choose a natural disaster and examine how it affects 6 regionally connected states.
2. We'll see how the natural disaster affects the population's risk based on size and income, and we'll use the risk definition as described by the NRI.
3. We will also look for regional trends within the data between the 6 chosen states.
4. We will be making it easy to read using tables and Summary plots to see how exactly the risk levels and populations are related.

Project Tasks:

1. Clean and organize data from an outside source
 - Our team will download the National Risk Index (NRI) Census Tract dataset from the FEMA data resources website, along with the NRI Data Dictionary and Hazard Information files. The CDC Social Vulnerability Index (SVI) dataset will also be obtained. All datasets will be reviewed in Python to confirm they are properly structured for analysis.

2. Filter data by 6 states and one Natural Disaster type
 - Maine, New Hampshire, Vermont, Massachusetts, Rhode Island, and Connecticut
 - Risk or winter weather

3. Summarize data for natural disaster of our choice
 - Using Python with the matplotlib and GeoPandas libraries, our team will create visualizations of tables and plots for each state and the region.

4. Analysis of the risk to population and regional trends
 - Using Python with the pandas and numpy libraries, our team will analyze the NRI dataset for each state. Summary statistics including mean, median, minimum, and maximum values will be calculated for key variables such as Expected Annual Loss, Social Vulnerability, Community Resilience, and NRI Risk Scores for winter weather. This analysis will help identify trends and differences in risk distribution across the region.

5. Compare high risks to population density
 - Using the data collected from our code using the risk score calculations, we will compare the high risk of winter weather to population density.

Deliverables:

The deliverables requested by the client, Risk Averse, LLC, are summarized in the following table:

Deliverable:	Description:	Submission Date:
Scope of Work	Project plan outlining goals, tasks, and methods	5/8/2026
Cleaned Data File (All states)	Original NRI and SVI datasets	5/8/2026
Filtered Data File (6 states)	Processed and merged datasets used in analysis	5/8/2026
Python Code	Jupyter Notebook containing all project code and plots and tables	5/8/2026
Annotated Code Document	Explanation of code functions and steps	5/8/2026
Final Report	Summary of analysis, results, and conclusions	5/8/2026
GitHub Repository	Organized project files and documentation	5/8/2026

